

An Explainable Data-Driven Platform for Optimal Wind Farm Siting

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1 Introduction

Wind farm site selection is a crucial yet highly complex challenge. Optimal siting involves much more than simply identifying windy locations; it requires balancing multiple, often competing factors including, but not limited to, proximity to the electricity grid, consistency of wind resources, land use constraints, environmental concerns, community acceptance, and economic viability [1–3]. The goal is to select locations that will not only generate the most reliable energy over time but also integrate well into the broader power system and surrounding landscape. However, this decision-making process is often not systematically optimized.

This issue has become even more pressing in the context of growing climate change awareness and the global transition toward sustainable energy systems. In Australia, policy commitments such as the Victorian Renewable Energy Targets, legislated in the *Renewable Energy Act (2017)* [4], reinforce this shift. These targets include:

- 40% of Victoria’s electricity generation to be renewable by 2025
- 65% by 2030
- 95% by 2035

Meeting such goals will demand not only increased renewable energy capacity, but also smarter, more coordinated deployment strategies. This underscores the need for data-driven and holistic planning tools that can guide decision-making.

Currently, wind farm development in Australia is largely driven by a bottom-up planning ap-

proach. While site selection occurs within a defined legal framework and requires approval from energy market regulators, final decisions are often shaped by the interests of local stakeholders, like landowners, and by wind resource assessments conducted over relatively short timescales. Given that wind farms typically operate for 20 to 30 years, this approach can result in long-term inefficiencies [5]. Studies have shown that such a fragmented planning process can lead to suboptimal outcomes in terms of both total energy generation and supply stability. In contrast, a top-down planning model, where wind farm locations are determined with long-term demand patterns in mind, has demonstrated the potential for significantly greater efficiency and reliability [6].

1.1 Aim

This literature review aims to explore existing methods for wind farm site selection, both globally and within Australia. It will examine traditional approaches as well as recent techniques that incorporate artificial intelligence, with a particular focus on how explainable these methods are, both in terms of technical transparency (e.g., explainable AI) and in their ability to clearly justify decisions to non-technical audiences.

Findings from the review will inform the design of a user-friendly, explainable decision-support platform intended for a diverse range of users. These can include both technical actors such as infrastructure planners, power operators (e.g., AEMO, VicGrid) and decision makers, as well as ordinary citizens and local communities. In this context, clear and interactive representations of site selection criteria can help build trust, reduce misunderstandings, and support more inclu-

sive planning. This is particularly important because the question of where to produce renewable electricity is often controversial, and the acceptance of new energy infrastructures, especially by the communities living near proposed sites, is critical, and it can lead to tension between stakeholders with different priorities or levels of technical understanding. [2, 7–9].



Figure 1: Recent protests against wind farms in Australia. **(Center)** A protest in Wollongong against plans for an off-shore wind farm, May 2025 (Photo: Anna Warr, The Advocate). **(Right)** A rally on the front lawns of Parliament House in Canberra, February 24 (Photo: Mike Bowers, The Guardian). **(Left)** A sign photographed by the author in Warrnambool, March 2025

1.2 Document structure

Following the introduction, this report presents a structured literature review divided into several sections. This review critically examines traditional and AI-based methods for wind farm site selection, with a particular focus on how explainable the methods are. It includes comparisons between approaches used across different countries, as well as a section dedicated to Australian visual and interactive tools, mainly focusing on mapping systems. These tools, coming from different domains than wind energy planning, offer inspiration and practical features that will help shape the design of the proposed platform. The State of the Art Summary then identifies gaps in the existing literature, particularly the lack of explainable and accessible visual tools that allow the creation of “what-if” scenarios. The Research Project Plan outlines a practical approach to addressing these gaps, including a section on the technical tools to be used, such as programming languages and frameworks. Finally, the Conclusion section highlights the key findings and provides a concise summary of the report.

2 Literature review

TODO

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